CLANCY CRAWFORD

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Aerospace Engineering & Computational Mathematics student with experience in requirements driving design and verification, structural/thermal analysis, and multidisciplinary engineering.

EDUCATION

Embry-Riddle Aeronautical University - Florida

- Bachelor of Science in Aerospace Engineering (May 2027)
  Rocket Propulsion Track
- Bachelor of Science in Computational Mathematics (May 2027)
  Engineering Application Track

John Wood Community College - Illinois

- Associates of Science - Engineering Science (May 2024)
  Collegiate Volleyball and High Honors (3.75 GPA)
- Associates of Science: *Minor in Physics* (Dec 2023)
  Collegiate Volleyball and Highest Honors (4.0 GPA)

EXPERIENCE

Mayott Aerospace — Connector Team Lead *Daytona Beach, FL | March 2026–Present*

- Lead connector team development for a heavy-lift drone payload docking system, focusing on reliable payload pickup, alignment tolerance, and mechanical interface design. Evaluated materials and weight reduction options for integration.
- Researched robotic tool changer mechanisms to guide connector geometry, load transfer, and self-aligning docking concepts.
- Supported company development efforts for Mayott Aerospace's Launch Your Venture competition, winning $10,000 in funding.

Mayott Aerospace — Operations Team Member *Daytona Beach, FL | March 2026–Present*

- Supported company organization and operational readiness by developing SOP processes, KPI tracking concepts, and audit preparation materials. Contributed to readiness documentation for funding, prototype planning, and internal team coordination.

PROJECTS – collaborated in multidisciplinary teams

Magnetic Alignment Dynamics for Drone Payload Docking Analysis May 2026

- Modeled a two-axis magnetic docking connector for a heavy-lift drone using nonlinear rotational differential equation dynamics.
- Delivered a technical report with derivations and MATLAB/ODE-based modeling to support connector design development.

Experimental Aerodynamics & Wind Tunnel Analysis Spring 2026

- Conducted wind tunnel experiments on airfoil, canard-delta wing, and supersonic nozzle configurations.
- Connected measured results in MATLAB to stall behavior, drag trends, shock formation and various flow behavior.

Causal Analysis of Aviation Accident Lethality May 2026

- Analyzed and converted fatal aviation accidents to identify high-lethality outcomes into cause categories using keywords.
- Applied FCI causal discovery, conditional independence, and regression causal effect estimation to evaluate relationships.

Avionics Box Structural & Thermal Analysis Design - SOLIDWORKS Apr 2025

- Modeled support and pipe structures; delivered technical reports and recommendations on material selection and design.
- Analyzed structural response under dynamic flight condition and performed structural analysis while meeting constraints.
- Conducted thermal & airflow analysis, developed performance criteria for temperature, and contributed to planning integration.

SKILLS

Programs: MATLAB, Simulink, SOLIDWORKS, CATIA, Fusion 360, LaTeX, MobaXterm, Femap
Technical: CAD, GD&T, structural and thermal analysis, wind tunnel data reduction, regression modeling
Languages: MATLAB, Python, Java, HTML, CSS **Certifications**: Microsoft Office: Word, PowerPoint, Excel

RESEARCH

Orbital Debris and Collision Avoidance in Low Earth Orbit – Presentation and Report

Researched orbital debris growth, tracking limitations, collision risk modeling, and debris mitigation strategies for spacecraft operating in Low Earth Orbit (LEO). Researched tools and frameworks within this field and presented research.

Numerical Methods for Ordinary Differential Equations – Presentation and Report

Implemented and compared numerical methods for solving ordinary differential equations, analyzing accuracy and stability. Applied numerical simulation techniques in MATLAB and python that can analyze dynamic accuracy and systems.

LEADERSHIP & INVOLVEMENT

ERAU: Club Volleyball Treasurer, Society of Women in Engineers (SWE), Society of Women in Space Exploration (SWSE), Embry Riddle Future Space Explorers and Developers Society (ERFSEDS)

JWCC: Collegiate Volleyball Athlete, Student-Athlete Advisory Council, Vice President of Phi Theta Kappa (PTK), Vice President of S.T.E.M. Club, Student Government Association

AWARDS

- Gold Scholar for Coca-Cola's Academic Team (National Level) May 2024
- All-USA Academic Team by Phi Theta Kappa Honor Society (National Level) May 2024
- Microsoft Office Specialist State Champion Microsoft Office Specialist (Word and Excel: Office 2016) May 2022
- Renewable Women of Excellence Scholarship ($5,000) Aug 2022